

WESTERN SYDNEY
UNIVERSITY



Module 6

Overview of Two-Dimensional Plotting – Part 1

Usefulness of plotting data

- Large sets of data are usually difficult to interpret as tables of numbers.
- Engineers and scientists use graphical techniques to reduce large sets of data to help gain insight.
- With graphical presentations insightful trends and possible errors are more easily identified.

plot function

- Creating (x, y) plots is easy with plot(), e.g.:

- Type two vectors:

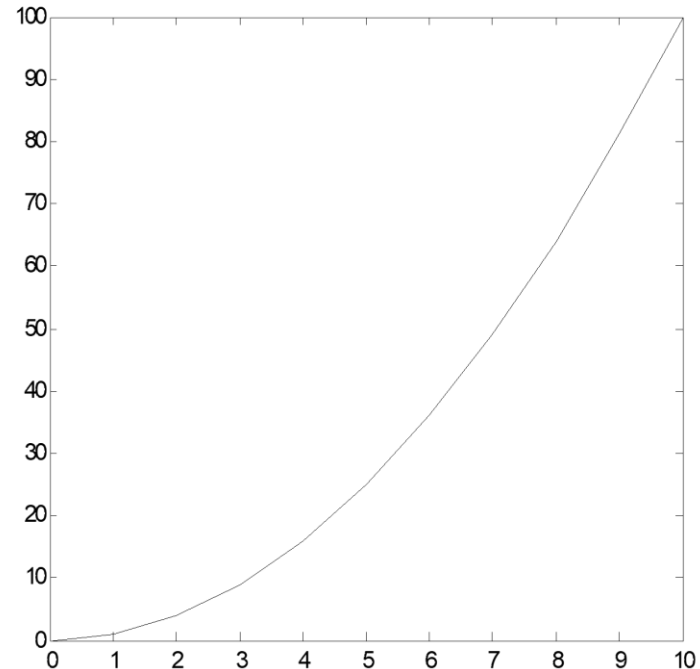
```
x = 0:1:10;
```

```
y = x.^2;
```

- Use the **plot()**

command:

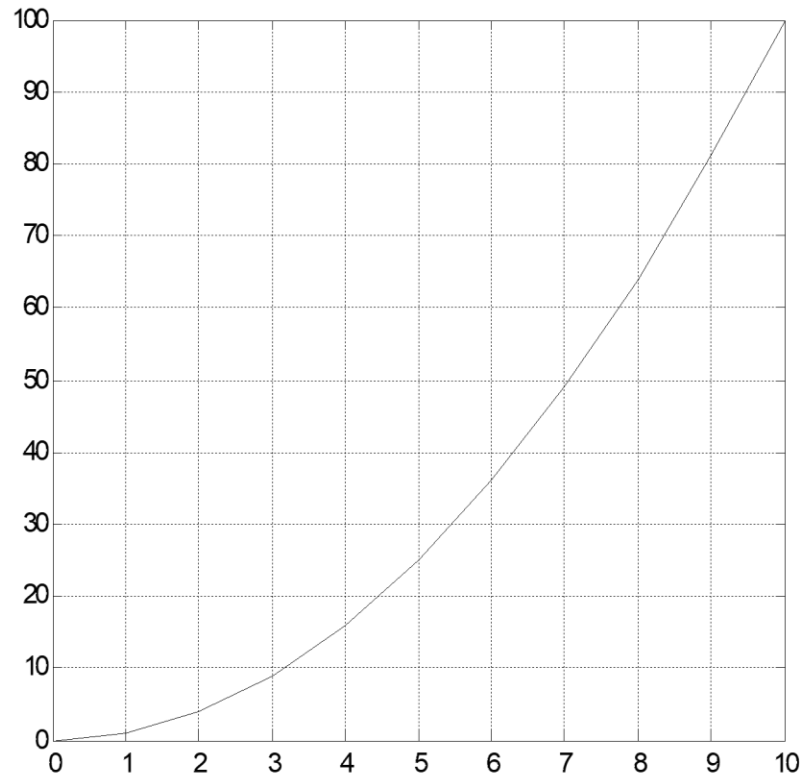
```
plot(x,y);
```



Add a grid

- Use the **grid** command to add a grid to the figure

```
grid on;
```



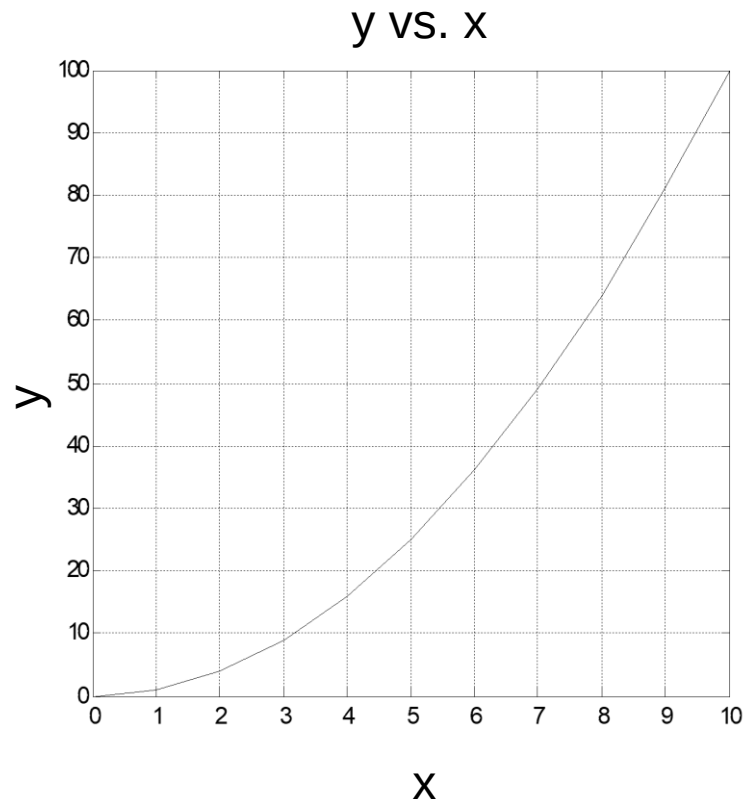
Add a title and labels

- Add a title and labels to a plot as illustrated in the example:

```
title('y vs. x');
```

```
xlabel('x');
```

```
ylabel('y');
```



Changing the line style

- The every third argument of a plot function can be used to change the line style (line specification)

`plot(x,y,'line spec')`

- The line specification is a string written as combination of 3 characters that control the colour, marker shape and line type
- The character combination can be selected via the table on the next page

Line specification table

Line style	Indicator	Marker Shape	Indicator	Colour	Indicator
solid	-	point	.	blue	b
dotted	:	circle	o	green	g
dash-dot	-.	x-mark	x	red	r
dashed	--	plus	+	cyan	c
		star	*	magenta	m
		square	s	yellow	y
		diamond	d	black	k
		triangle down	v		
		triangle up	^		
		triangle left	<		
		triangle right	>		
		pentagram	p		
		hexagram	h		

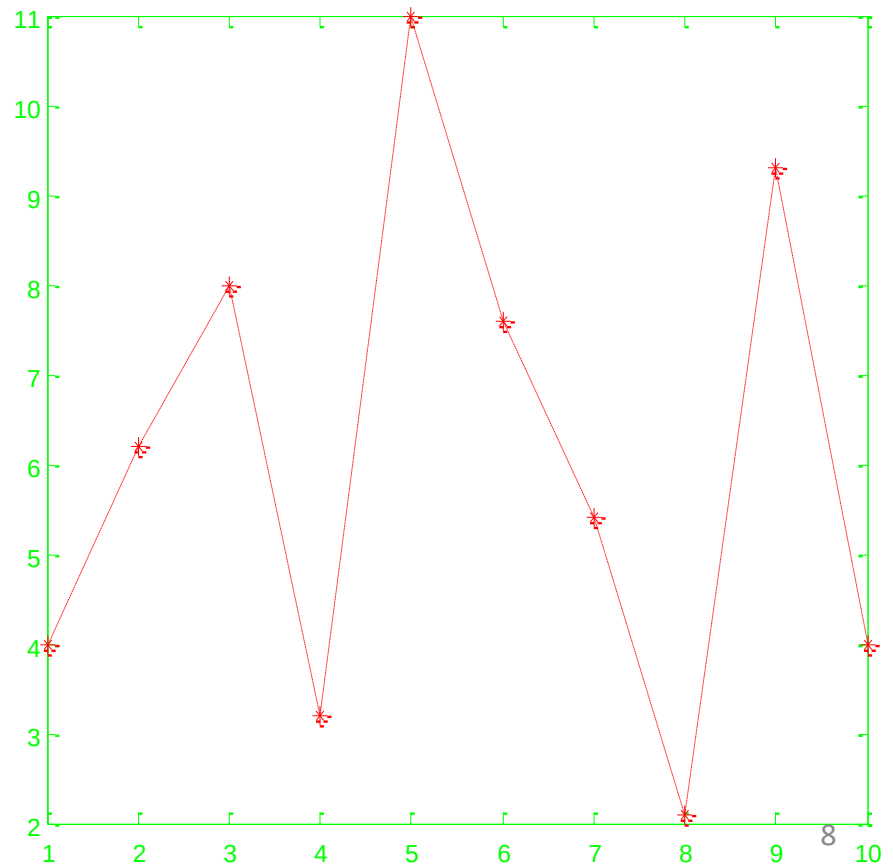
Example of changing line style

```
x = 1:1:10;
```

```
y = [4, 6.2, 8, 3.2, 11, 7.6, 5.4, 2.1, 9.3, 4];
```

- To plot the data with a red, dash-dot line with red stars for points:

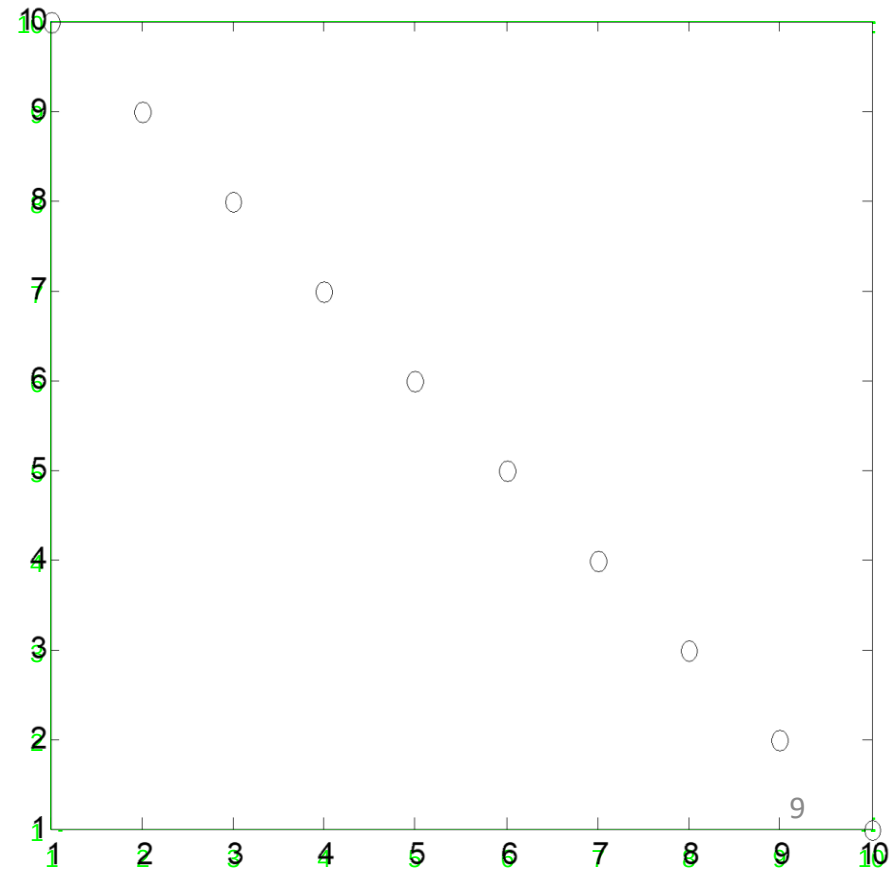
```
plot(x,y,'r-.*');
```



Plotting individual data points

- Setting a marker type without specifying a line type will suppress the straight line drawn by default between the points that define the lines.

```
x = 1:1:10  
y = 10:-1:1  
plot(x,y)  
plot(x,y, 'o')
```



Plots of complex numbers

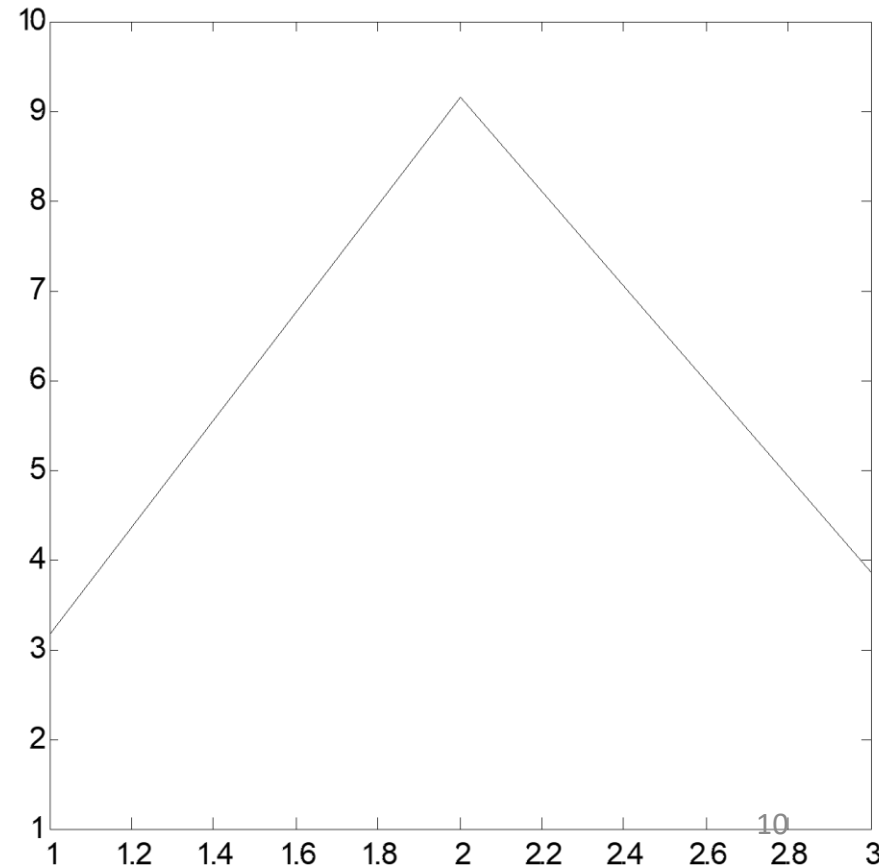
- If input to the plot function is complex, MATLAB plots the real component on the x-axis and the imaginary on the y-axis. Alternatively, if

```
A = [1 + 1i, 1 + 2i,  
      2 + 3i, 3 + 4i];
```

```
B = sin(A);
```

```
plot(A,B);
```

- Note: MATLAB gives a warning that imaginary parts are ignored.



Re-scaling the axis

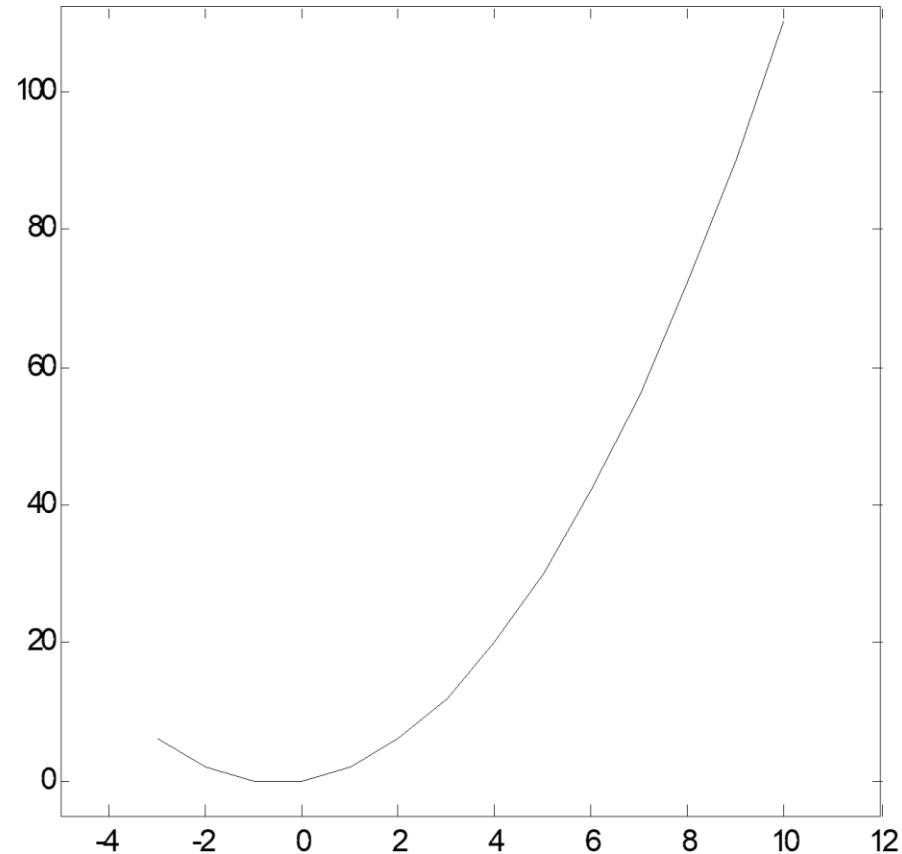
```
x = -3:1:10;
```

```
y = x.^2 + x;
```

```
plot(x,y) ;
```

■ If you do not like the fact that the curve touches the axis, you can change the scale as follows:

```
axis([-5, 12, -5, 112]);
```



Annotations with figures

```
x = 1:1:10;  
y = x.^2;  
plot(x,y);  
text(2,90,'Add text to  
plots with the  
text command.');
```

